

Bayesian Modeling of Dose–Response Curves: Estimating Trametinib IC_{50} in Cancer Cell Lines

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1 Introduction

Precision oncology relies on predicting how specific cancer profiles respond to targeted therapies. In this project, I analyze *in vitro* drug screening data to evaluate the dose–response dynamics of **Trametinib**, a targeted cancer drug widely used in the treatment of several malignancies. The pharmacological measurements and the associated biological annotations were gathered from specialized public databases, primarily the *Cancer Dependency Map (DepMap)* Portal maintained by the Broad Institute.

The primary objective of this study is to propose a fully Bayesian analysis for estimating the half-maximal inhibitory concentration (IC_{50}) — the exact drug concentration required to reduce cancer cell viability by 50%. Rather than relying on a single point estimate, the project leverages Markov Chain Monte Carlo (MCMC) simulation to naturally incorporate uncertainty and extract full posterior distributions for the parameter of interest, providing a mathematically robust inferential framework for nonlinear dose–response models affected by biological variability and experimental noise.

The Hill Equation

To model the relationship between drug concentration and biological response, pharmacology relies on the Hill equation, often generalized as the four-parameter log-logistic model (LL.4). In its standard form, the model reads

$$E = E_{min} + \frac{E_{max} - E_{min}}{1 + \left(\frac{IC_{50}}{Dose}\right)^h}.$$

where

- E denotes the observed biological response (cell viability);
- $Dose$ is the administered drug concentration;
- E_{max} and E_{min} represent the upper and lower asymptotes of the response curve;
- IC_{50} denotes the concentration producing a 50% reduction of the maximal response;
- h is the Hill coefficient, which controls the steepness of the dose–response curve.

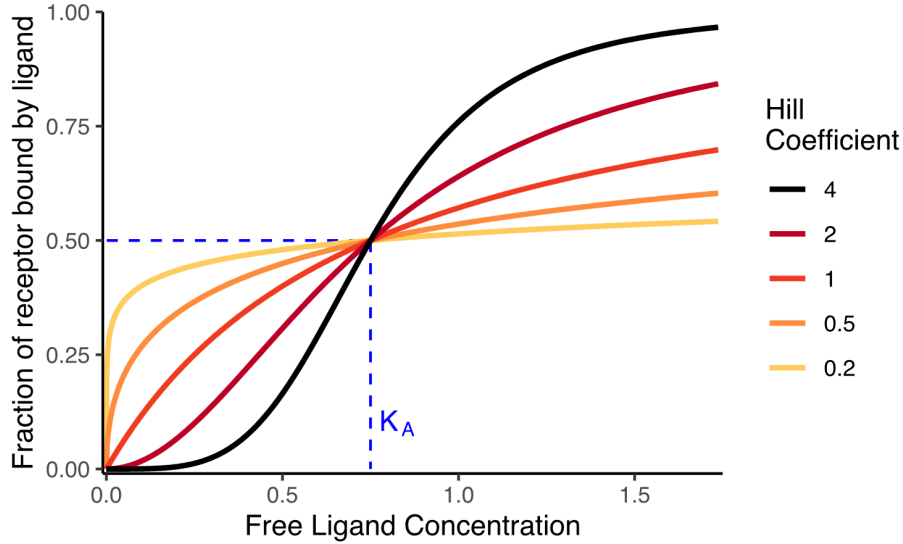


Figure 1: Standard Hill equation curves illustrating different values of the Hill coefficient (h). In its classical formulation, the biological response increases with increasing drug concentration.

As illustrated in Figure 1, the classical Hill equation is predominantly used to model an **increasing** effect — for instance, how effectively a treatment cures a symptom, how well an antibiotic induces rest, or the growing fraction of receptor bound by a ligand.

In the present study, however, I am evaluating Trametinib on the basis of its toxicity to cancer cells, so the outcome of interest is **cell viability** (survival) rather than efficacy. Because of this, the model behaves essentially “in reverse”: as the drug concentration increases, cell survival decreases. The dose–response curves therefore describe an inhibitory dynamic, starting from maximum viability at low doses (E_{max}) and dropping toward complete cell death at high doses (E_{min}). The formulation is adapted accordingly, subtracting the drug’s effect from the maximum baseline:

$$E = E_{max} - \frac{E_{max} - E_{min}}{1 + \left(\frac{IC_{50}}{Dose}\right)^h}.$$

2 Data Collection and Dataset Engineering

To build a robust Bayesian model, the raw pharmacological data required rigorous cleaning and structuring in order to isolate the true biological signal from experimental noise. The initial dataset contained high-throughput viability measurements for hundreds of cancer cell lines exposed to multiple drug concentrations.

The preprocessing workflow followed two main steps.

1. Drug Selection

The dataset was first filtered to retain only experiments involving **Trametinib**. The resulting subset included multiple cancer cell lines, each evaluated over a range of drug concentrations, allowing the complete dose-response profile to be observed.

2. Genomic Filtering

A critical step in precision-oncology modeling is making sure that the observed biological response is truly driven by the drug’s primary mechanism of action, rather than by overlapping genetic variables. To this end, the pharmacological measurements were cross-referenced with somatic mutation data, and cell lines presenting known off-target mutations were excluded, in order to better isolate the pharmacological response associated with the mutation of interest.

Ideally, one would additionally exclude cell lines genetically predisposed to acquire these confounding alterations. However, this would require a separate analysis based on genomic predisposition data that are generally unavailable in publicly accessible datasets. Consequently, the present filtering strategy should be regarded as a practical approximation rather than a complete characterization of all potential biological confounders.

3 Statistical Modeling Framework

The IC_{50} parameter was estimated through a fully Bayesian framework implemented in JAGS. The MCMC sampler used a standard, robust configuration: three parallel chains, each consisting of 20,000 iterations, with the first 5,000 iterations discarded as burn-in.

3.1 Parameter Isolation via Frequentist Pre-computation

In order to sharply focus the Bayesian inference on the primary biological parameter of interest, I adopted a parameter-isolation strategy. Rather than estimating all four parameters of the LL.4 equation simultaneously—which would drastically increase the posterior dimensionality and computational complexity—preliminary frequentist fits were performed using the **drc** package in R.

These preliminary models estimated the upper asymptote ($E_{\max}$), the lower asymptote ($E_{\min}$), and the Hill coefficient (h). These estimates were then supplied to the Bayesian model as fixed inputs. This strategic dimensionality reduction allows the MCMC sampler to concentrate all of its resolving power entirely on exploring the posterior distribution of the IC_{50} .

3.2 Model 1: Generic Pooled Model

As a baseline specification, I first considered a pooled model that assumes a common drug sensitivity across all cell lines. Under this assumption, a single global IC_{50} parameter is estimated from the entire dataset.

```
1 jags_data_generic <- list(  
2   viability = data$viability,  
3   dose = data$dose,  
4   N = nrow(data),  
5   h_fixed = fixed_params$generic_h_reference[1],  
6   min_fixed = 0,  
7   max_fixed = 1  
8 )  
9  
10 jags_model_generic <- "  
11 model {  
12  
13   for (i in 1:N) {  
14  
15     viability[i] ~ dnorm(mu[i], tau)  
16  
17     mu[i] <- min_fixed +  
18       (max_fixed - min_fixed) /  
19       (1 + pow(dose[i] / IC50, h_fixed))  
20  
21   }  
22  
23   IC50 ~ dgamma(0.001, 0.001)  
24  
25   tau ~ dgamma(0.001, 0.001)  
26  
27 }  
28 "
```

Listing 1: JAGS Script: Generic Pooled Model

Observed viability values are modeled through a Normal likelihood. As specified by the $viability[i] \sim dnorm(mu[i], tau)$ statement in the model code, the expected value μ_i is evaluated dynamically for each individual data point based on the administered dose and the theoretical LL.4 curve. However, the precision term τ is estimated as a single global parameter shared across all observations. This reflects the standard statistical assumption of **homoscedasticity**: the observational noise (experimental variance) is assumed to remain constant across the entire dose range, regardless of the viability level.

Weakly informative Gamma priors are assigned to both the global IC_{50} parameter and the precision term τ , enforcing positivity while leaving the posterior distribution

essentially data-driven.

Because the LL.4 model enters the likelihood through a nonlinear transformation, the Gamma prior is not conjugate here, meaning the resulting posterior distribution has no closed-form solution. This mathematical intractability is precisely why MCMC sampling is required to approximate the posterior space numerically.

3.3 Model 2: Hierarchical Patient-Specific Model

Given the strong biological heterogeneity among cell lines, the pooled assumption is clearly too rigid. I therefore built a hierarchical (multi-level) model that estimates a distinct IC_{50} for each cell line, while still allowing information to be shared across the population (partial pooling).

```
1 jags_data_patient <- list(  
2   viability = data$viability,  
3   dose = data$dose,  
4   patient = data$patient_idx,  
5   N = nrow(data),  
6   n_patients = nrow(fixed_params),  
7   h_fixed = fixed_params$h_slope,  
8   min_fixed = fixed_params$min_val,  
9   max_fixed = fixed_params$max_val  
10 )  
11  
12 jags_model_patient <- "  
13 model {  
14  
15   for (i in 1:N) {  
16  
17     viability[i] ~ dnorm(mu[i], tau)  
18  
19     mu[i] <- min_fixed[patient[i]] +  
20               (max_fixed[patient[i]] -  
21               min_fixed[patient[i]]) /  
22               (1 + pow(dose[i] /  
23               IC50[patient[i]],  
24               h_fixed[patient[i]]))  
25  
26   }  
27  
28   for (p in 1:n_patients) {  
29  
30     IC50[p] ~ dgamma(shape_ic50,  
31                    rate_ic50)  
32  
33   }
```

```

34
35   shape_ic50 ~ dgamma(0.01, 0.01)
36
37   rate_ic50 ~ dgamma(0.01, 0.01)
38
39   tau ~ dgamma(0.001, 0.001)
40
41 }
42 "

```

Listing 2: JAGS Script: Patient-Specific Hierarchical Model

Each cell line’s drug sensitivity is drawn from a shared, population-level distribution:

$$IC_{50,p} \sim \text{Gamma}(\text{shape_ic50}, \text{rate_ic50}). \quad (1)$$

The hyperparameters governing this population distribution are themselves assigned prior distributions and estimated from the data. This hierarchical structure induces partial pooling: well-measured cell lines collectively establish the population norm, while a regularization effect commonly referred to as *shrinkage* pulls unstable or noisy individual estimates toward that population mean — without forcing artificial homogeneity across genuinely different cancer lines.

4 Model Comparison

4.1 Deviance Information Criterion (DIC)

To compare the competing model specifications, I employed the Deviance Information Criterion (DIC),

$$DIC = \bar{D} + p_D, \quad (2)$$

where $\bar{D}$ represents the posterior mean deviance and p_D denotes the effective number of parameters. Lower DIC values indicate a more favorable balance between model fit and model complexity.

The pooled and hierarchical models produced the following results:

Model Comparison Results

- Generic Pooled Model DIC: **162.0198**
- Hierarchical Model DIC: **-1884.843**

In principle, the p_D term works against the hierarchical model, since it estimates a distinct IC_{50} for every cell line. In practice, however, the pooled model fits the highly

heterogeneous biological data so poorly that its residual deviance ($\bar{D}$) is enormous — so much so that the improvement in goodness-of-fit achieved by the hierarchical architecture completely absorbs the added complexity penalty, producing a drastic reduction in DIC.

These findings are consistent with the biological expectation that drug sensitivity varies across cancer cell lines and cannot be adequately represented by a single population-wide parameter.

4.2 Posterior Predictive Checks

To further assess model adequacy, I performed Posterior Predictive Checks (PPCs) by comparing the observed viability measurements with replicated datasets generated from the posterior predictive distribution (Figure 2).

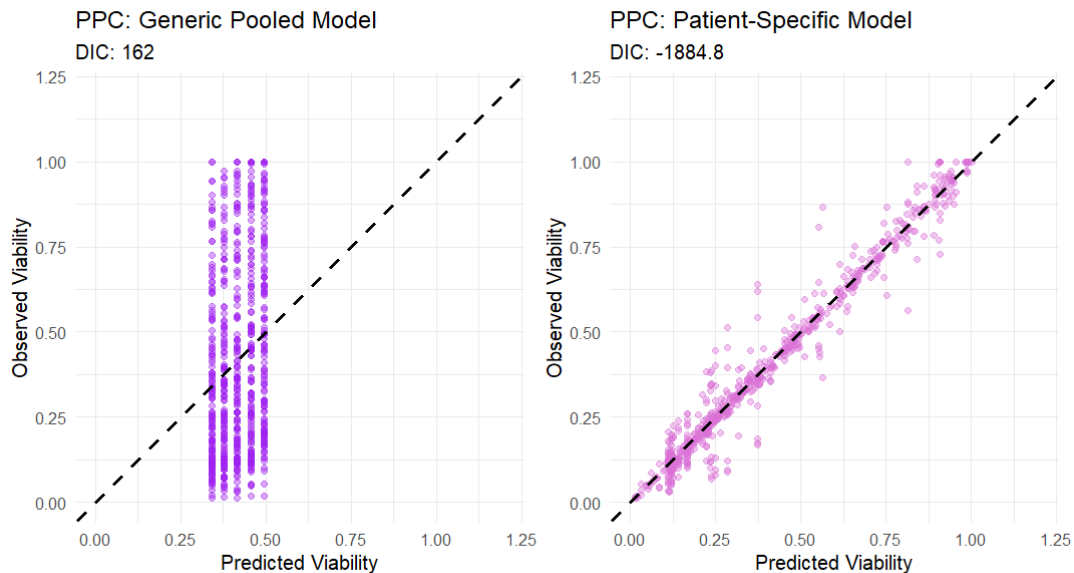


Figure 2: Posterior Predictive Checks comparing observed and posterior-predicted viabilities for the Generic Pooled Model (left) and the Hierarchical Model (right).

The two model specifications exhibit markedly different predictive behaviour.

- **Generic Pooled Model.** The predicted values occupy a relatively narrow range and fail to reproduce the observed variability across cell lines, reflecting the restrictive assumption of a common IC_{50} shared by all observations.
- **Hierarchical Model.** The posterior predictions cluster much closer to the identity line ($y = x$), indicating substantially better agreement between observed and predicted viabilities. Allowing cell-line-specific parameters while borrowing information across the population lets the model capture both individual variability and overall population structure.

The posterior predictive checks therefore support the conclusions obtained from the DIC analysis, confirming that the hierarchical specification provides a considerably better description of the observed data.

5 Parameter Recovery

Before applying the hierarchical model to real biological data, I evaluated its ability to recover known parameter values through a simulation study. Parameter recovery is an important validation step: it verifies that the estimation procedure can accurately infer the parameters used to generate a synthetic dataset.

I generated a synthetic dose–response profile using a known parameter configuration with

$$IC_{50} = 0.08, \quad E_{min} = 0.2, \quad E_{max} = 1, \quad h = 1.5,$$

Dose concentrations were simulated across the experimental range, random observational noise was added to the generated responses, and the hierarchical Bayesian model was fitted using the same MCMC configuration adopted for the real dataset.

The recovery results are summarized below.

Parameter Recovery Results

- True IC_{50} : **0.0800**
- Posterior mean: **0.0819**
- 95% credible interval: **0.0576 – 0.1147**

The posterior mean closely matches the true parameter value used during simulation, and the 95% credible interval comfortably contains it. These results indicate that, under controlled conditions, the proposed Bayesian implementation is capable of accurately recovering the parameter of interest, with no evidence of systematic estimation bias.

6 Bayesian Estimation and Empirical Observations

6.1 Posterior Distributions and Empirical Estimates

To illustrate the additional information provided by Bayesian inference, I compared posterior distributions with empirical estimates obtained directly from the observed dose–response profiles.

Bayesian Posterior Densities vs. Empirical Observations

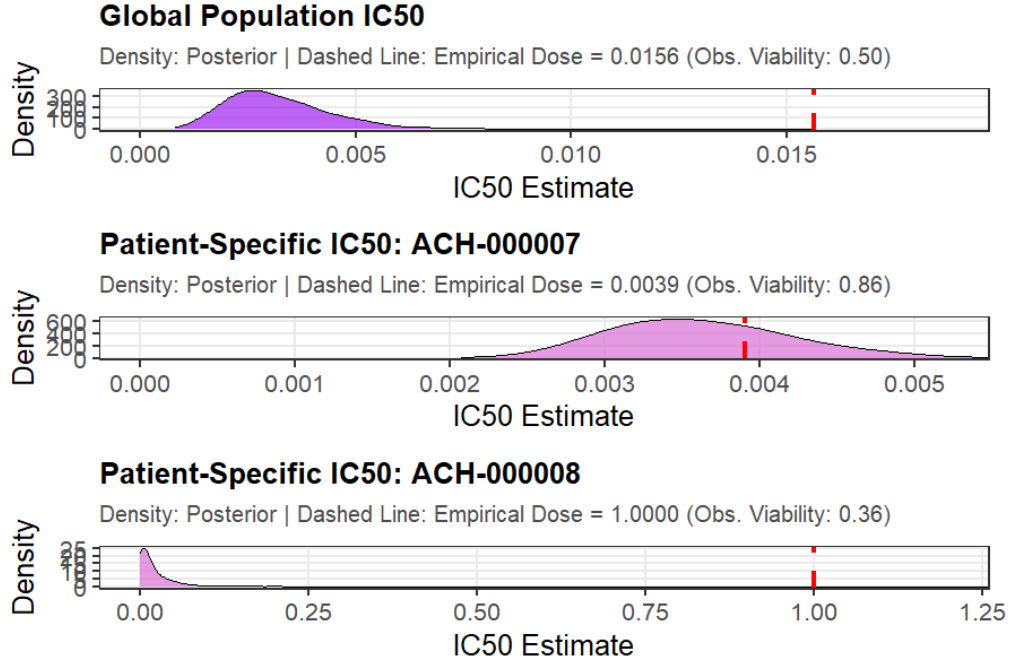


Figure 3: Posterior distributions for the global population and representative cell lines. Dashed vertical lines indicate empirical estimates obtained from the observed dose–response curves.

A naive empirical estimate of IC_{50} is often obtained by identifying the dose at which viability sits halfway between its minimum and maximum. For well-behaved dose–response profiles, this empirical estimate is generally located close to the centre of the posterior distribution — but relying on a single point or a crude midpoint approximation is inherently fragile: the Bayesian approach additionally quantifies estimation uncertainty by providing the complete posterior distribution, rather than a single point estimate.

6.2 Robustness to Noisy Observations

The advantages of Bayesian inference become more evident when the observed data contain substantial experimental variability or isolated outliers.

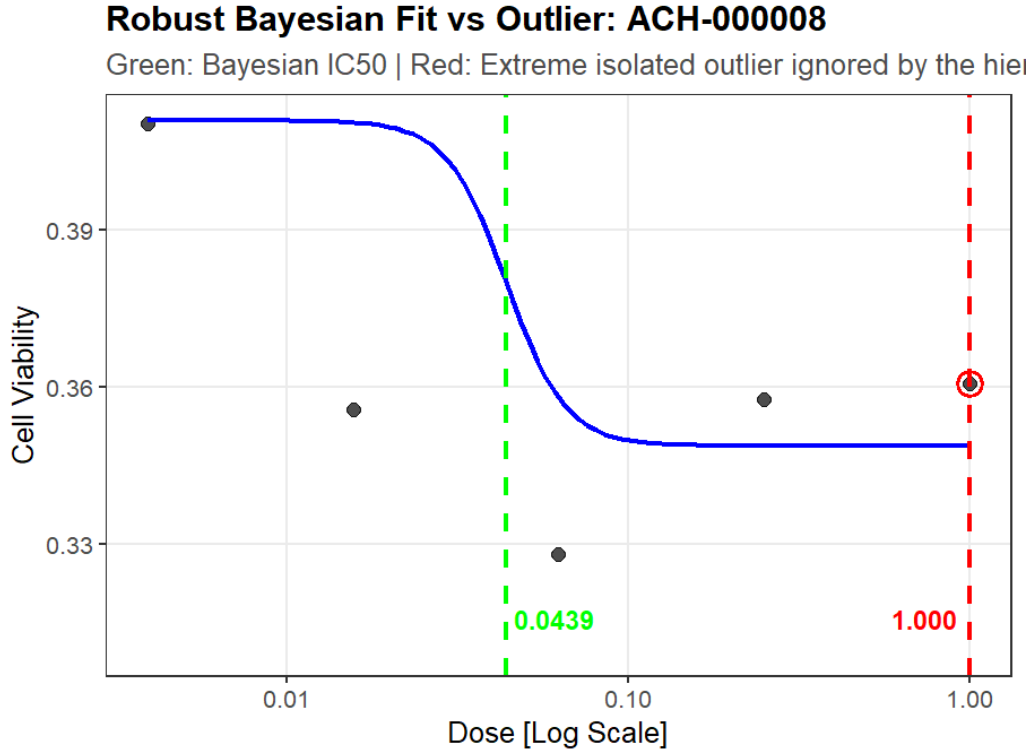


Figure 4: Hierarchical Bayesian fit for cell line ACH-000008. The estimated dose–response curve is only moderately influenced by an isolated observation because inference is based on the complete likelihood and the hierarchical population structure.

Unlike empirical procedures that may rely heavily on a small number of observations, Bayesian inference combines information across the entire dose–response profile. Within the hierarchical framework, individual estimates are further stabilized through partial pooling, reducing the influence of isolated noisy measurements while remaining consistent with the overall data pattern. Consequently, the estimated IC_{50} remains compatible with the complete set of observations rather than being driven by a single, potentially influential point.

7 Bayesian and Frequentist Estimation

7.1 Point Estimates versus Posterior Distributions

Classical nonlinear regression methods typically summarize inference through a single point estimate for each parameter. Bayesian inference, in contrast, represents uncertainty explicitly by estimating the complete posterior distribution.

Bayesian Posterior Densities vs. Frequentist Estimates

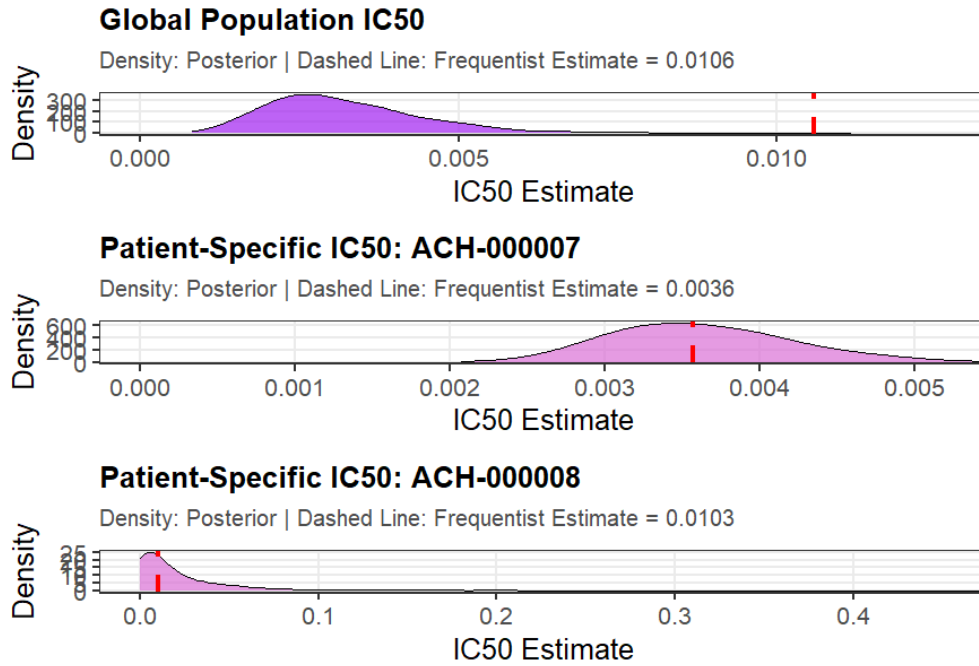


Figure 5: Comparison between Bayesian posterior distributions and frequentist point estimates (dashed red lines) for the global population and representative cell lines.

Figure 5 shows that, for well-behaved dose-response profiles, frequentist point estimates are generally located close to the posterior mode. Nevertheless, in a high-stakes field such as oncology, carrying forward only a single “best guess” is fundamentally limiting: the Bayesian framework additionally provides a direct quantification of estimation uncertainty through credible intervals and posterior probability distributions, ensuring that critical medical decisions are not based on a single, fragile number.

7.2 Shrinkage in the Hierarchical Model

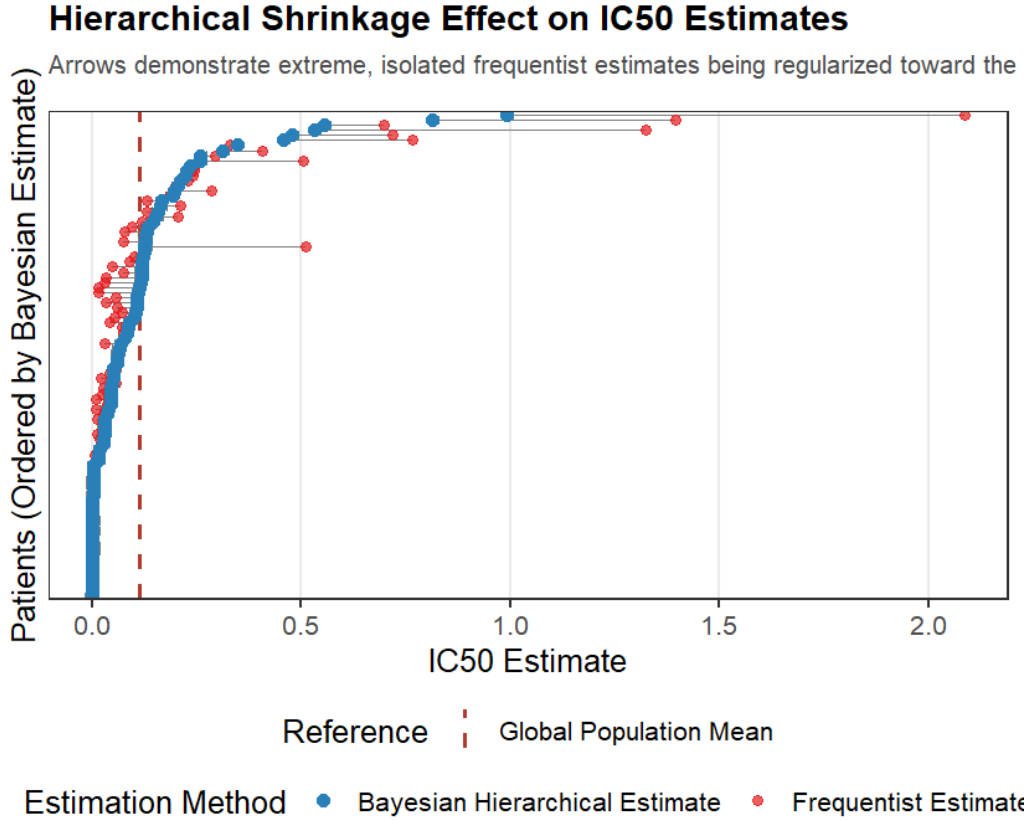


Figure 6: Comparison between hierarchical Bayesian estimates (blue) and frequentist estimates (red) across all cell lines. The dashed vertical line indicates the population-level mean.

Figure 6 illustrates the effect of hierarchical modeling on the estimated IC_{50} values. Frequentist estimates are obtained independently for each cell line and therefore depend exclusively on the local information available for that individual profile; estimates from noisy or sparsely sampled dose-response curves can display substantial variability as a result.

The hierarchical Bayesian model partially pools information across all cell lines by assuming that individual IC_{50} values arise from a common population distribution. This induces a regularization effect — *shrinkage* — whereby estimates based on limited information are moderately pulled toward the population mean, while well-supported estimates remain largely determined by their own data. This reduces estimation variability without eliminating biologically meaningful differences among individual cell lines.

8 Sensitivity Analysis and Posterior Learning

8.1 Prior Sensitivity Analysis

I conducted a sensitivity analysis to evaluate the influence of the prior distribution on posterior inference, considering three alternative Gamma prior specifications for the IC_{50} parameter.

Posterior Mean Estimates

- Vague prior (Gamma(0.001, 0.001)): **0.0673**
- Weakly informative prior (Gamma(1, 1)): **0.0770**
- Strong informative prior (Gamma(10, 10)): **0.5594**

The posterior estimates obtained under the vague and weakly informative priors are very similar, indicating that the observed data provide enough information to dominate these relatively diffuse prior specifications. The strongly informative prior, in contrast, produces a substantially different posterior estimate.

This behaviour reflects a fundamental characteristic of the experimental setup: **data sparsity**. Because each individual dose–response profile contains only a limited number of observations, the likelihood does not carry enough statistical weight to overwhelm a highly informative prior — which then acts as a flood of pseudo-observations that drowns out the sparse, real data. Overall, the sensitivity analysis confirms the necessity of a vague hierarchical prior structure, while showing that excessively informative priors can dominate the likelihood when the available data are relatively sparse.

8.2 Prior and Posterior Distributions

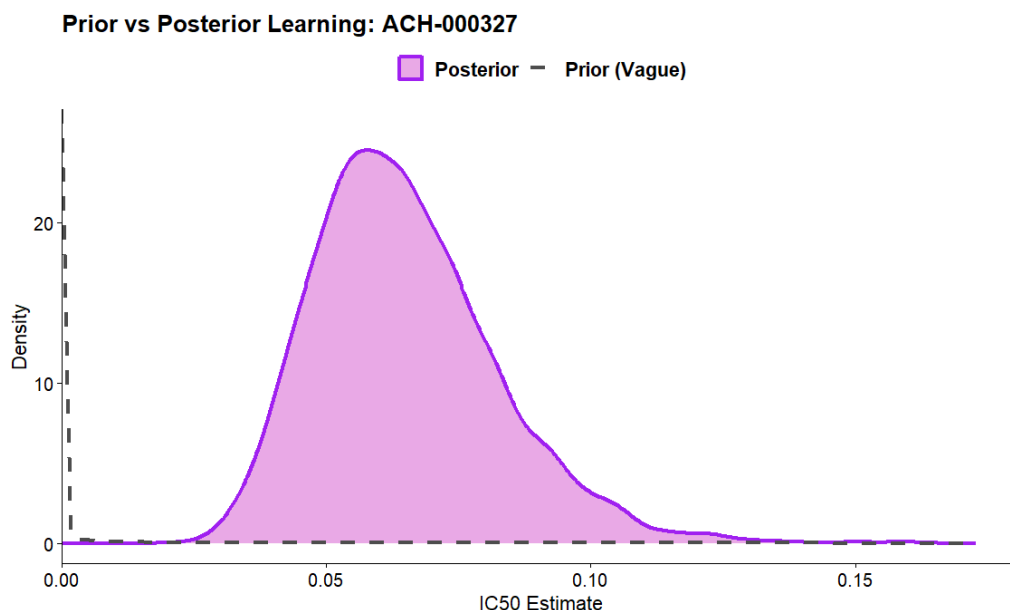


Figure 7: Comparison between the prior distribution and the posterior distribution for cell line ACH-000327.

Figure 7 illustrates how the information contained in the observed data updates the initial prior distribution. While the prior is intentionally diffuse, the posterior distribution becomes substantially more concentrated after incorporating the experimental observations, reflecting the information gained through Bayesian updating.

8.3 Posterior Predictive Dose–Response Curve

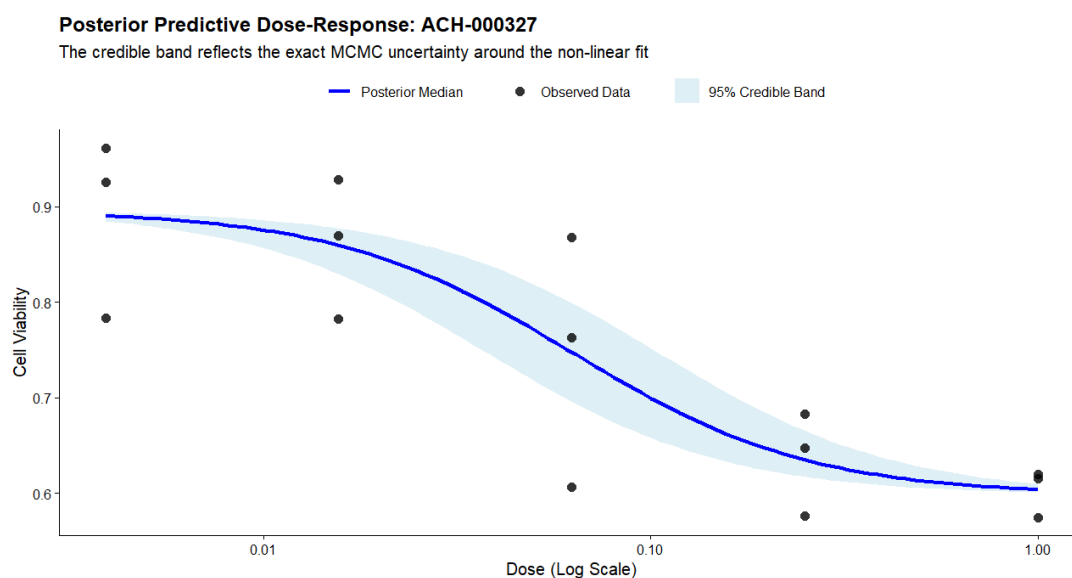


Figure 8: Posterior predictive dose–response curve for cell line ACH-000327. The shaded region represents the 95% credible interval.

The posterior predictive curve shown in Figure 8 summarizes the estimated dose–response relationship after integrating uncertainty over the posterior distribution of the model parameters. The posterior median follows the observed viability measurements closely, while the 95% credible interval reflects the uncertainty associated with the estimated nonlinear relationship across the range of administered doses.

9 MCMC Convergence and Formal Diagnostics

To guarantee the reliability of the posterior samples generated by the sampler, I evaluated both visual and formal statistical diagnostics for a representative cell line (ACH-000327).

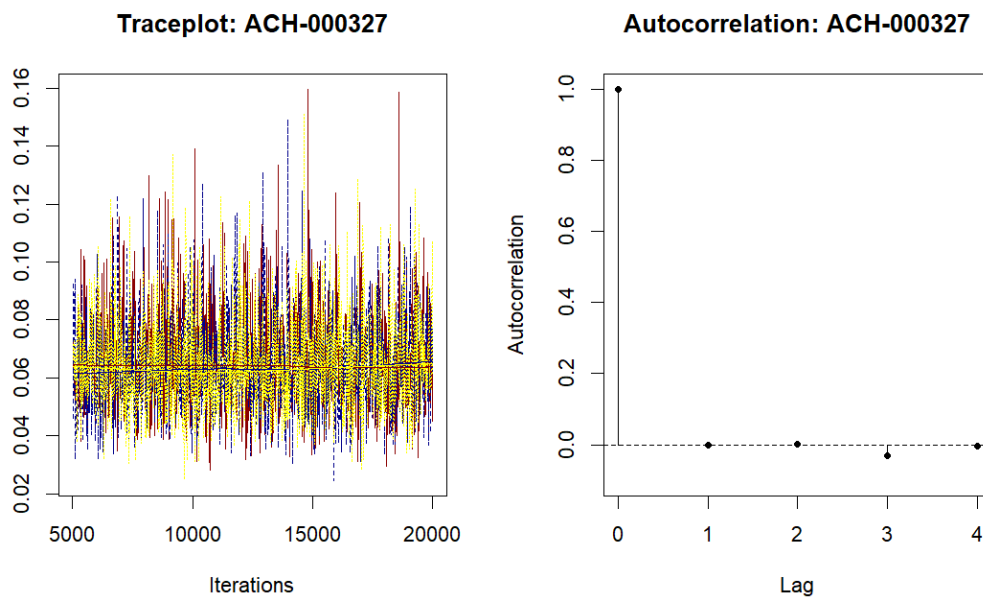


Figure 9: Traceplot (left) and Autocorrelation plot (right) demonstrating stationarity and sample independence for the IC_{50} parameter of ACH-000327.

Visual inspection of the traceplot (Figure 9, left panel) reveals the typical “fuzzy caterpillar” behaviour: the multiple parallel chains mix rapidly and fluctuate steadily around a common stationary distribution, without long-term trends, indicating good convergence. The autocorrelation plot (right panel) shows that correlation between consecutive samples drops to zero immediately, confirming that the chains are efficiently exploring the parameter space and producing independent samples.

To quantify this, I also computed formal diagnostic statistics:

Formal MCMC Diagnostics

- Gelman–Rubin ($\hat{R}$): **1.0014** (Target: ≈ 1.0)
- Effective Sample Size (n_{eff}): **2300**

The Gelman–Rubin potential scale reduction factor ($\hat{R}$) sits practically at the target value of 1.0, showing that between-chain and within-chain variance are equivalent. Combined with a high effective sample size, these metrics confirm that the MCMC sampling is statistically sound, efficient, and well calibrated.

10 Future Directions and Possible Improvements

While the current hierarchical Bayesian framework successfully addresses the challenges of IC_{50} estimation, a few avenues for future refinement remain open.

- **Fully Bayesian 4-parameter estimation.** The current architecture treats IC_{50} as the only stochastic node, while keeping the remaining components of the LL.4 curve fixed. A natural evolution is a fully Bayesian 4-parameter hierarchical model, assigning a Gamma prior to the (positive) Hill slope h and Beta priors to the minimum and maximum asymptotes E_{min} and E_{max} — naturally bounded in $[0, 1]$ as viability proportions. This would let the model capture the full multidimensional uncertainty of the dose–response profile.
- **Genomic covariates and mutational filtering.** Biological heterogeneity is heavily driven by underlying genetic profiles: some cell lines may show high survival not because the drug is broadly ineffective, but because they carry pre-existing resistance mutations. Future iterations could dynamically filter out cell lines with known confounding mutations or, more elegantly, incorporate genomic markers as covariates in the hierarchical structure, preventing resistant subpopulations from skewing the baseline efficacy estimate for the rest of the cohort.

References and Data Sources

- **Figure 1 and Hill equation:**
Wikipedia contributors. *Hill equation (biochemistry)*. [https://en.wikipedia.org/wiki/Hill_equation_\(biochemistry\)](https://en.wikipedia.org/wiki/Hill_equation_(biochemistry))
- **Data:**
Broad Institute. *Cancer Dependency Map (DepMap) Public Dataset*. https://depmap.org/portal/data_page/?tab=allData
- **Software and Frequentist Pre-computation:**
Ritz, C., Baty, F., Bollen, J. C., & Walley, J. (2015). *Dose-Response Analysis Using R*. PLOS ONE, 10(12), e0146021. <https://doi.org/10.1371/journal.pone.0146021>